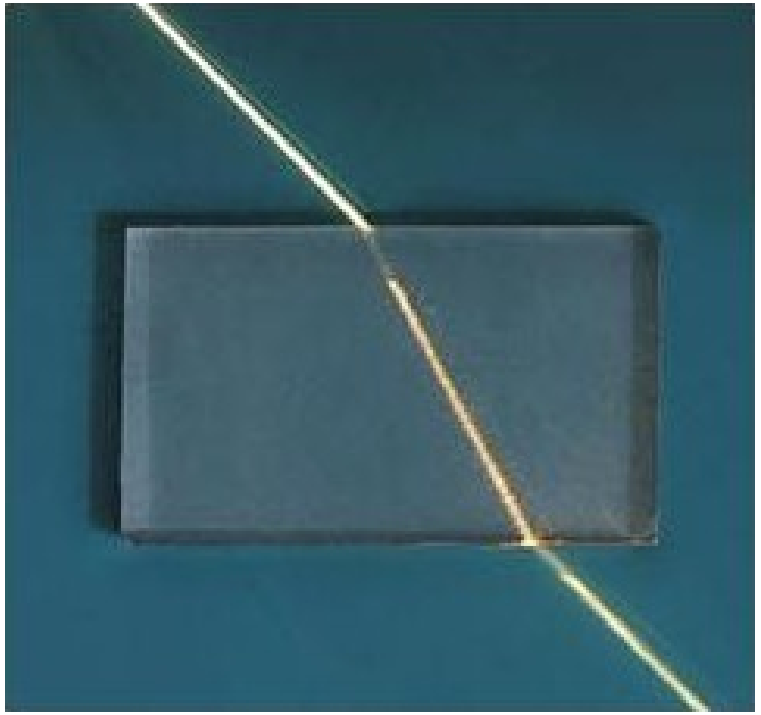
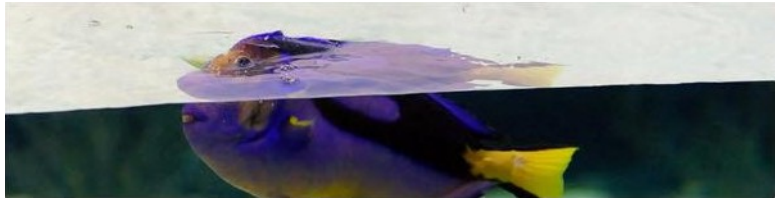
4.04 Wave Speed and Density: Refraction

Why does light bend when it enters a new medium?





Think About It



Why does this polar bear look like its head is chopped off?

Make your best guess. We will find out today!

Single choice poll

A. It is really two different bears.

Single choice poll

B. Light bends when it goes between water and air.

Single choice poll

C. The water acts like a mirror.

Single choice poll

D. The bear is actually shaped like that.

Today's Goals

I can define refraction: a wave bends when it enters a new medium.

I can explain that the wave's speed changes, so it bends.

I can tell when light bends toward or away from the normal.

I can use PhET data to show light slows down in glass and water.

Rate each goal:

1 Not yet

2 Getting there

3 I can teach it

Ohio Science Standards

7.PS.4 Energy can be transferred through a variety of ways.

- A wave travels at a constant speed through a material as long as the material is uniform.
- The speed of a wave depends on the material (sound travels faster through most solids than gases).
- When a wave enters a new medium, its speed changes, which can make it bend (refraction).

Inquiry skills (Grades 5–8): "Use appropriate mathematics, tools and techniques to gather data" and "Think critically and logically to connect evidence and explanations" (today's Refraction Lab).

Source: Ohio's Learning Standards for Science (2018), Grade 7 Physical Science; content statement paraphrased.

In this lesson

- Integer warm-up
- Vocab check-in
- What is refraction?
- Toward or away from the normal
- The marching band model
- Refraction Lab (graded)
- Lenses and our eyes

MISSION: FLIP THE SIGN!

Subtracting integers • Add the opposite

The Integer Crew is trapped! Recode subtraction as addition to unlock the gate.

1 KEEP the first. 2 CHANGE to +. 3 FLIP the second!

SUBTRACT A POSITIVE

The second number flips to negative.



$7 - 10 = 7 + (-10) = -3$

SUBTRACT A NEGATIVE

The second number flips to positive.



$-9 - (-2) = -9 + 2 = -7$

FIND THE CHANGE: end − start → $6 - 11 = -5$

Negative change = down or left.

A math model: robots represent signed numbers.

CASE #1 • MONDAY

How much warmer is the gym?

WAVE WORD: *Wave speed*
How fast a wave travels through a medium.

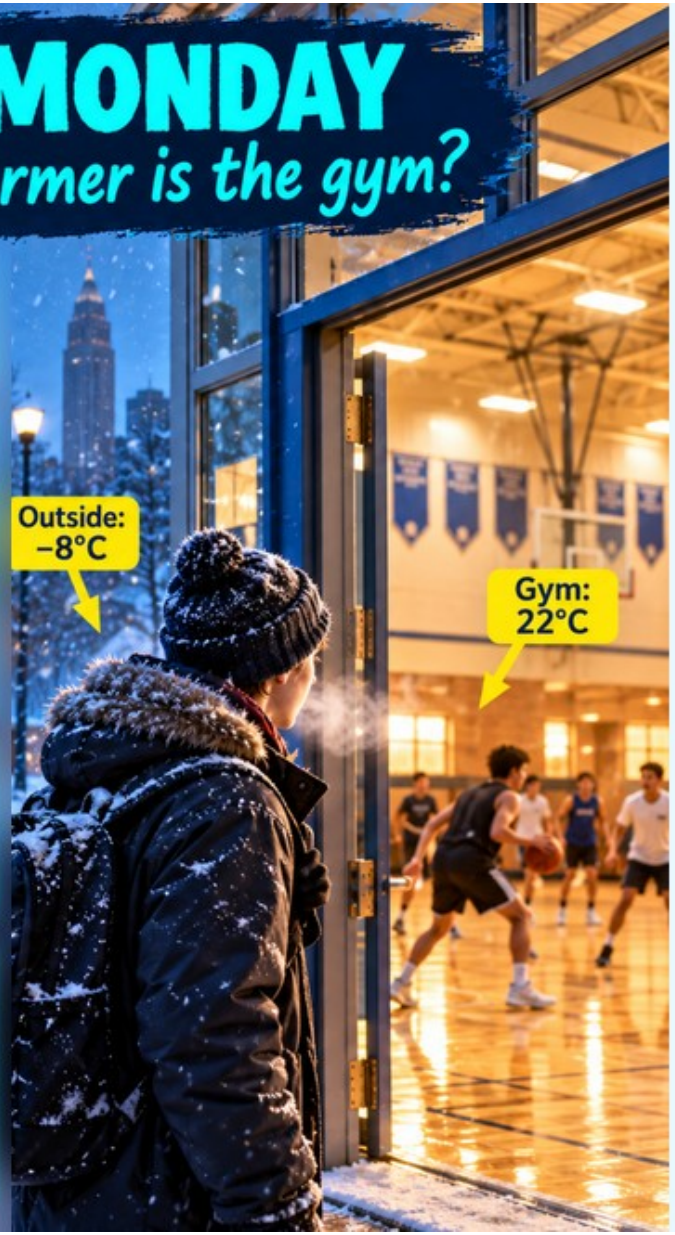
MYSTERY

Sound waves travel faster in warm air than in cold, denser air. On a January morning in Cleveland, it is 8 degrees below zero (Celsius) outside. Inside the school gym, it is 22°C. A student says the gym is only 14 degrees warmer. Is the student right?

CLUES

- Outside: −8°C
- Gym: 22°C
- Student says: 14°





CLAIM

Is the student right?

☐ YES ☐ NO

EVIDENCE

Solve:

22 − (−8) = _____

Rewrite as adding: 22 + _____ = _____

REASONING

Finish the sentence:

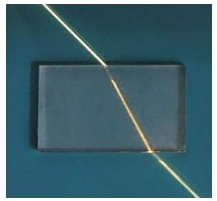
22 − (−8) is the same as

22 + _____ = _____.

So the gym is _____ degrees warmer.

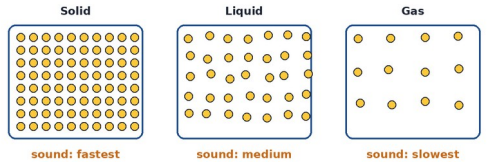
Vocab Check-In

From your Unit 4 vocabulary pretest. Today we focus on the starred words.



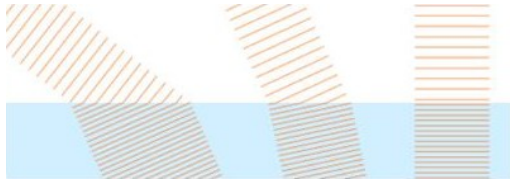
refraction ★

the bending of a wave when it enters a different medium



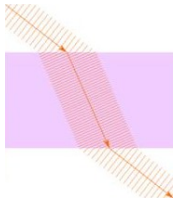
density ★

how much mass is packed into a space; glass and water are denser than air



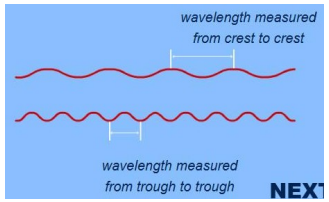
medium ★

the material a wave travels through (air, water, glass)



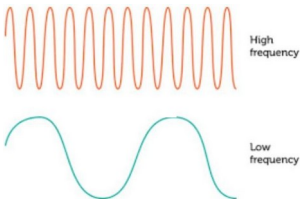
normal

an imaginary line at a right angle (90°) to the surface



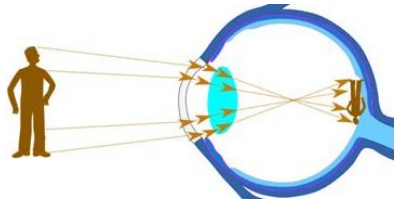
wavelength ★

the distance from one crest to the next crest



frequency ★

the number of waves that pass a point each second



lens

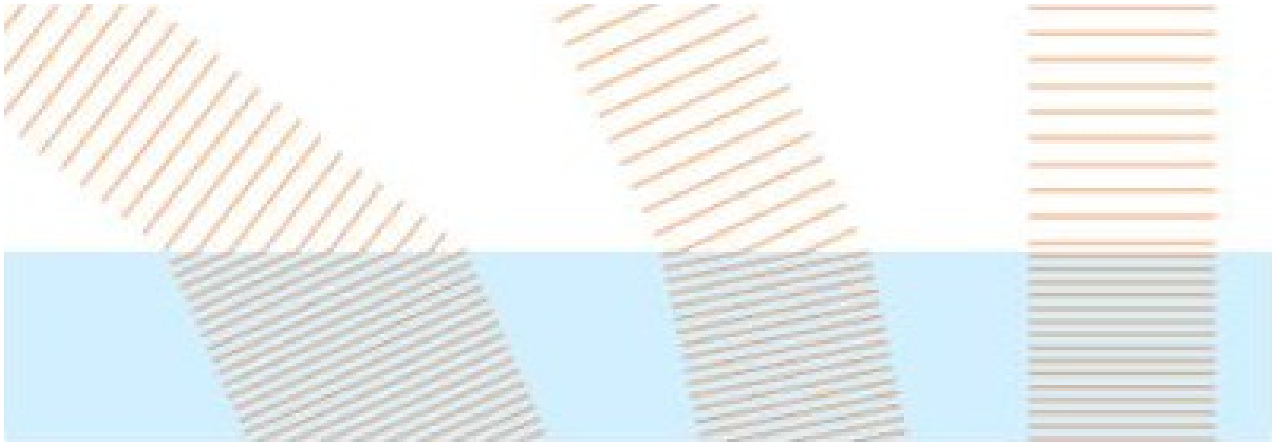
a curved, clear object that bends light to focus it

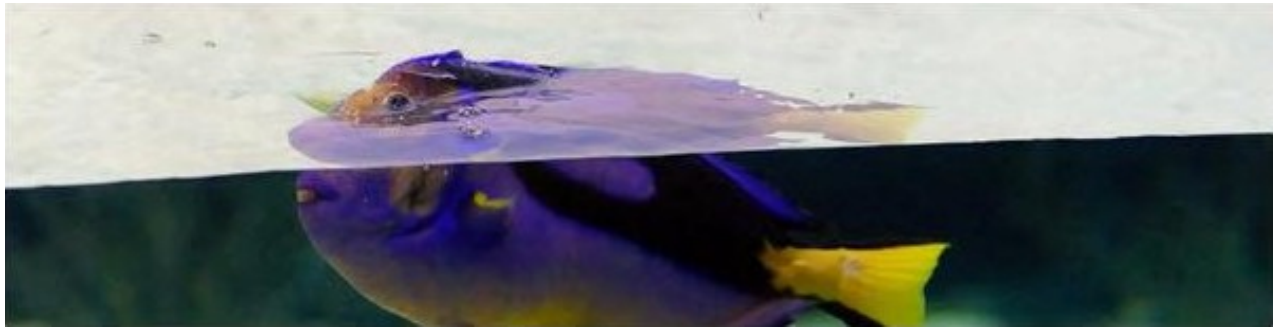


wave speed

how far a wave travels each second; it depends on the medium

What Is Refraction?





Refraction = the bending of a wave when it enters a different medium.

In the new medium:
speed **changes**
frequency **stays the same**
wavelength **changes**
direction **bends (if it hits at an angle)**

Pictures from the 4.04 page: the pulses bunch up in the water, and the fish looks shifted at the waterline.

Watch: Refraction



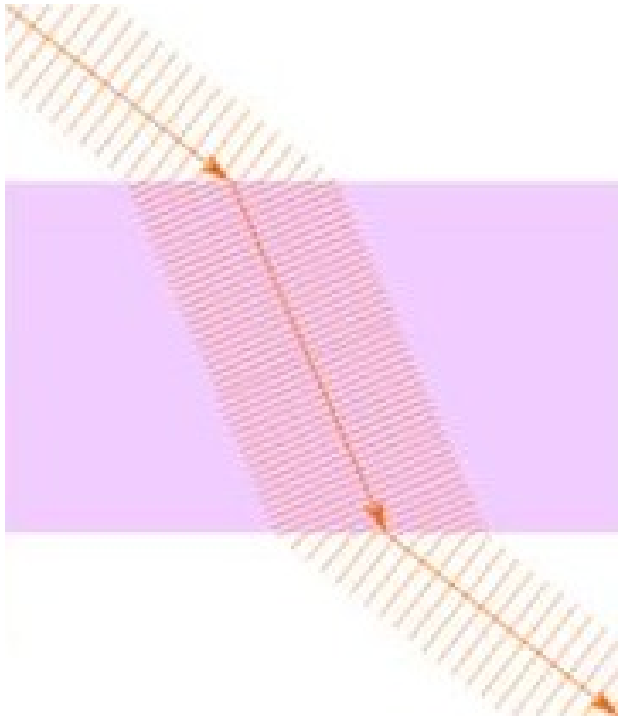
Refraction video from the 4.04 page (plays here in Slide Show)

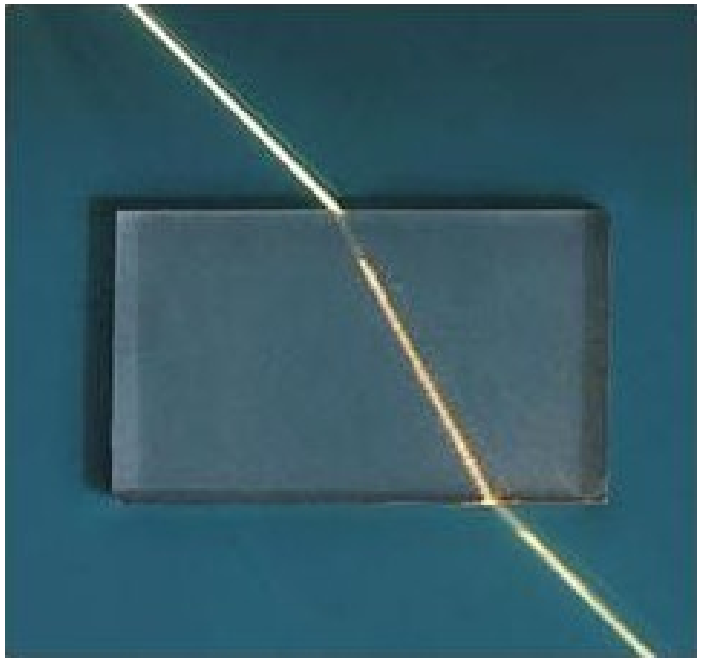
Watch the red ray:

- Air into glass: toward or away from the normal?
- Glass into air: toward or away?
- Straight in: does it bend?

▶️ Backup: YouTube

Toward or Away From the Normal?





Into a DENSER medium

light slows down and bends TOWARD the normal

Into a LESS dense medium

light speeds up and bends AWAY from the normal

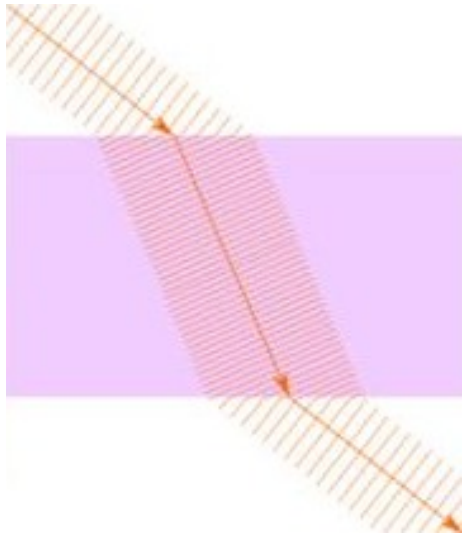
Straight in (90°)

no bending, but the speed still changes

Normal: an imaginary line at a right angle (90°) to the surface. Angles are measured from the normal.

Engageli Poll

Classkick: 4.04 Part 1 · Q1, Q3



Light shines at an angle from the air into a glass block. Glass is denser than air. What happens to the light?

Single choice poll

A. Speeds up, bends toward the normal

Single choice poll

B. Slows down, bends toward the normal

Single choice poll

C. Slows down, bends away from the normal

Single choice poll

D. Same speed, goes straight

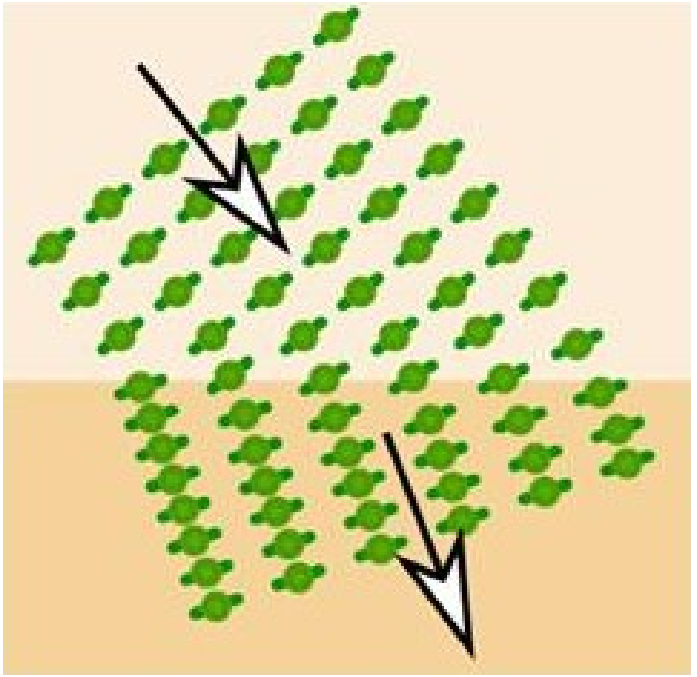
Classkick: 4.04 Part 1 · Q2, Q4

Why Does It Bend? The Marching Band

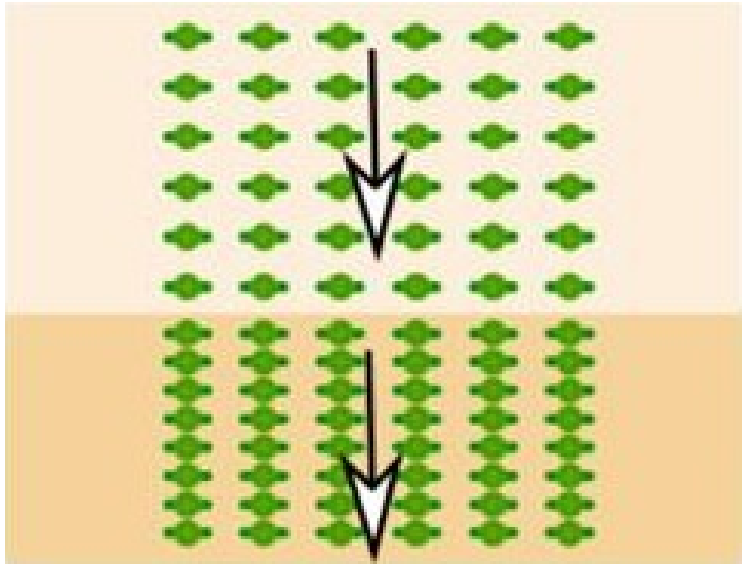


Flat ground: fast

Light does the same thing when it enters a denser medium.



At an angle: one side slows first, so the line turns.



Straight on: everyone slows together, no turn.

Who Is Right?

A straw in a glass of water looks broken right at the water line. Who explains it best?

Ava says:

"The water pushes on the straw and really bends it."

Ben says:

"Light bounces off the water like a mirror."

Cam says:

"Light changes speed between water and air, so it bends."

Dee says:

"The water soaks up the light, so part of the straw disappears."



Single choice poll

A. Ava



Single choice poll

B. Ben



Single choice poll

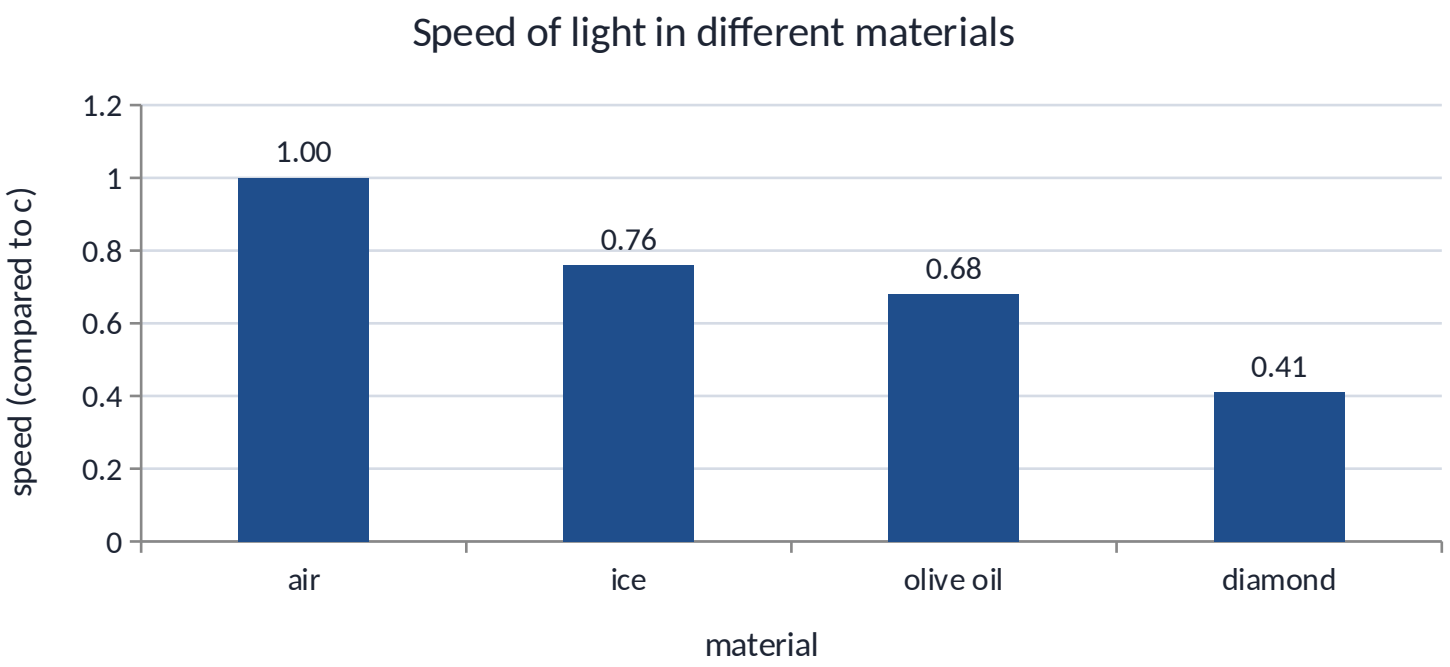
C. Cam



Single choice poll

D. Dee

Graph Poll



Read the graph.

A light ray goes from air into each material at an angle. In which one does it bend MOST?

Single choice poll

A. ice

Single choice poll

B. olive oil

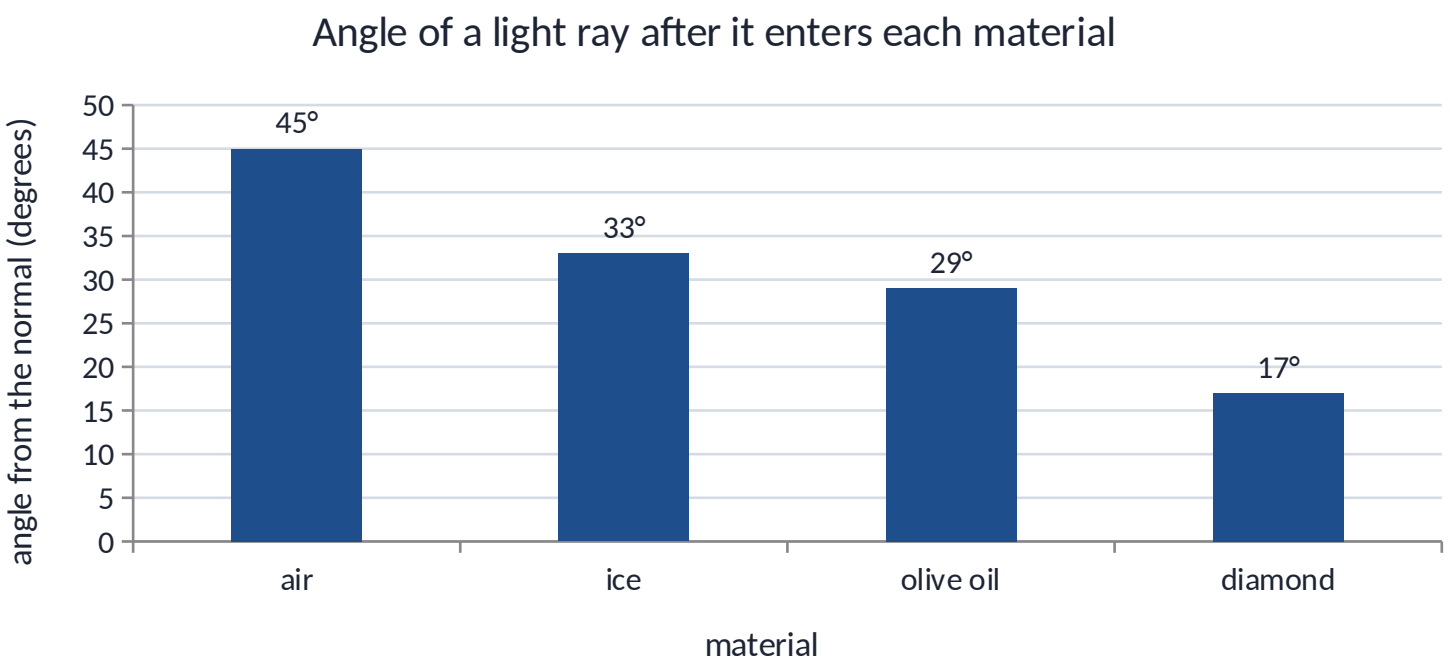
Single choice poll

C. diamond

Single choice poll

D. all the same

Graph Poll



Read the graph.

The ray started at 45°. What pattern does the graph show?

Single choice poll

A. Slower material, bigger angle

Single choice poll

B. Slower material, closer to the normal

Single choice poll

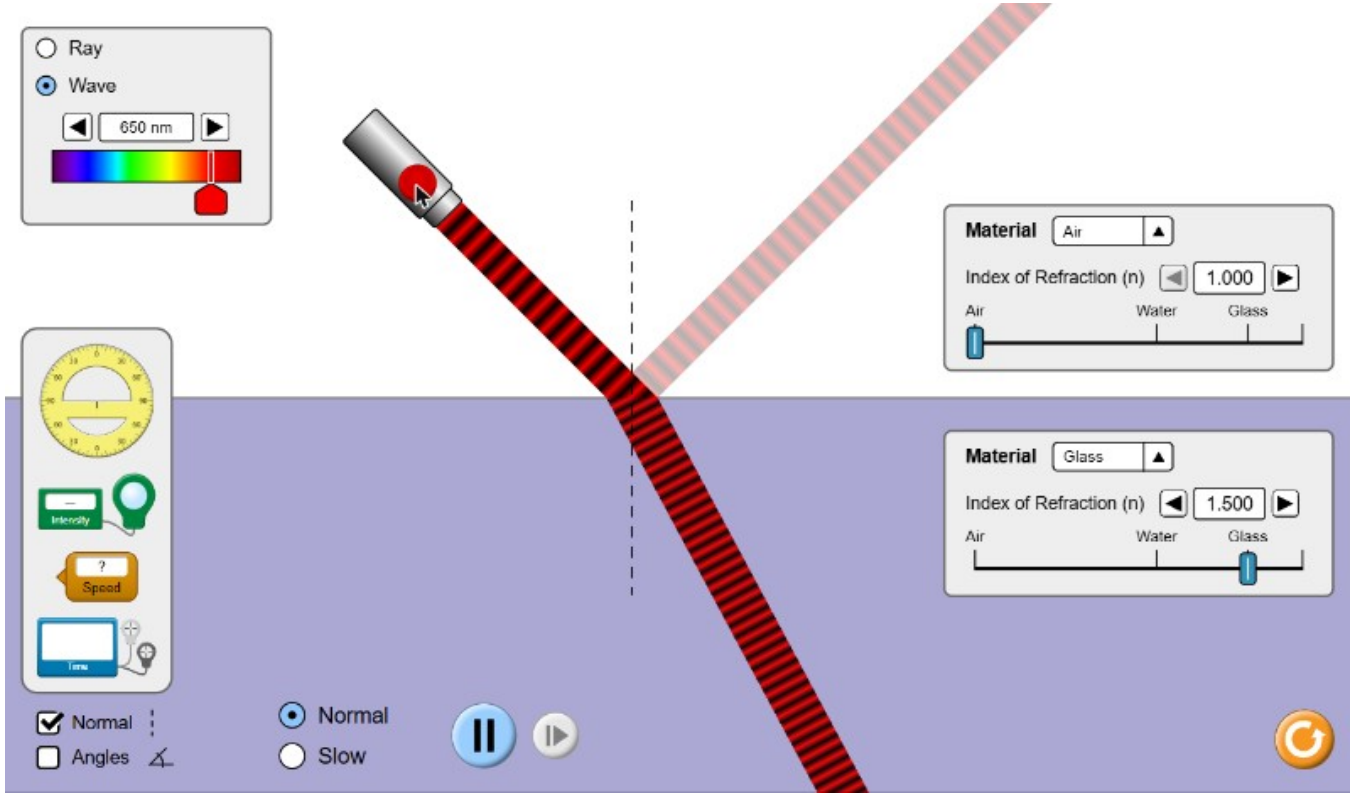
C. The angle never changes

Single choice poll

D. Faster material, more bending

Classkick: 4.04 Lab • Q5–Q7

Refraction Lab: PhET Bending Light



1. Open More Tools. Switch Ray to Wave. Set 650 nm.
2. Press the red button to turn on the laser.
3. Set the top and bottom materials: AIR / AIR, then AIR / GLASS, then AIR / WATER.
4. Put the Speed tool on the top beam, then the bottom beam. Record the speeds.

▶️ **Open PhET Bending Light**

The speed tool compares speed to c, the speed of light in empty space.

Classkick: 4.04 Refraction Lab Map

Get Ready for the Refraction Lab

Air → Air

Same medium on both sides: no change in speed and no bending.

Air → Glass

Glass is denser, so light slows down and bends toward the normal.

Don't fall for it

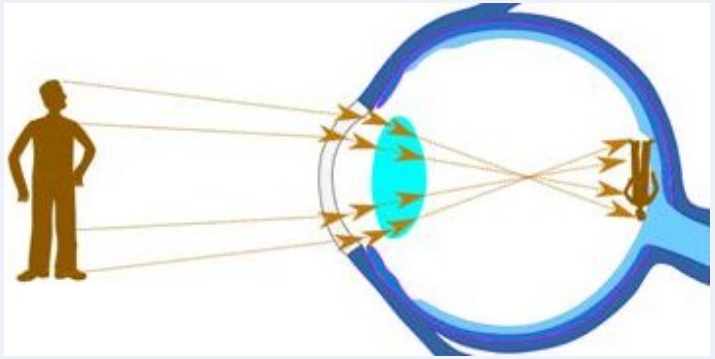
Light does NOT speed up in glass. Denser medium = slower light = bending.

Watch out!

Light is slower in denser materials. Sound is the opposite: faster in solids.

4.04 Refraction Lab (10 pts): assigned today. Due Mon, Nov 23 (extended time: Mon, Nov 30). Test each answer in PhET!

Refraction in Real Life



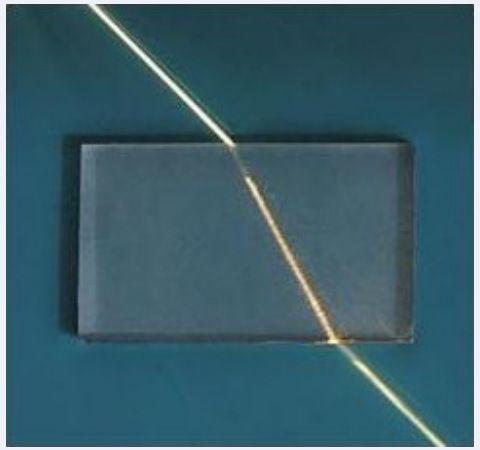
Your eye

The lens bends light to focus it on the back of the eye. The image is upside down, and your brain flips it!



Looking up from underwater

The whole sky squeezes into a bright circle at the surface.



A plastic block

Light bends going in and bends back coming out.

Key Ideas

1. Refraction = a wave bends when it enters a new medium.
2. The speed changes. The frequency stays the same, so the wavelength changes.
3. Into a denser medium, light slows down and bends toward the normal.
4. Light is slower in denser materials. Sound is faster in solids.

Re-rate your goals: 1 = Not yet, 2 = Getting there, 3 = I can teach it